John Benjamín Castro Sanabria

AI Engineer | Fintech & Agentic AI

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Portfolio | LinkedIn | GitHub | X

Professional Summary

AI Engineer in Novacomp's AI Office, with 4 years of combined experience in full-stack development, regulated fintech and Microsoft Cloud pre-sales. I build autonomous agents and solutions with LangGraph, RAG, MCP, Python and FastAPI, and AI-assisted software in C#/.NET, Java and Python. I have built payment, authentication and banking systems with C#/.NET, React, Angular and SQL; reduced regression from 3 days to 6 hours and a critical query from 47 s to 3.1 s.

Technical Skills

Development: C#, .NET / ASP.NET Core, Java, Spring Boot, Python, Flask, FastAPI, Node.js / Express, React, Angular, Vue, Blazor, TypeScript, JavaScript, React Native

Data and architecture: SQL Server, T-SQL, PostgreSQL, Oracle, MySQL, MongoDB, REST APIs, Clean Architecture, JWT, Entity Framework Core

Delivery and quality: Azure DevOps, Azure App Service, GitHub Actions, Docker, Git, xUnit, NUnit, MSTest, Moq, Playwright, Selenium

Applied AI: LangGraph, RAG, MCP, Langfuse, Ollama, FastAPI, observability, guardrails, Docker Sandboxes, Claude, Codex

Professional Experience

AI Engineer

Novacomp | November 2026 - Present

- Member of Novacomp's AI Office, working closely with engineers, architects, clients and product owners on client and internal requests.
- Build AI-assisted software in C#/.NET, Java and Python with Docker, using specifications, coding agents and human review.
- Design autonomous agents and LLM solutions with LangGraph, RAG and MCP.
- Build a technical assessment platform with six AI agents that draft questions, grade answers and analyze integrity, on .NET 10, SQL Server and React, with Microsoft Agent Framework, Claude models and OpenTelemetry traces.

Software Engineer

Innovative S.A. (Novacomp Group) | April 2026 - November 2026

- Developed and maintained enterprise systems for financial-sector clients across Java, .NET, React and SQL Server, with TypeScript/JavaScript front ends and Python tooling.
- Reduced regression from 3 days to 6 hours with 22 critical-path E2E suites in Playwright and Selenium and parallel xUnit/NUnit tests.
- Caught 11 defects before production across 4 consecutive releases of a 24/7 payments platform.
- Integrated LLM APIs and Python scripts into applied-AI workflows for financial report generation.

SDR / Pre-Sales Engineer

Servicios Computacionales Novacomp S.A. | October 2025 - March 2026

- Generated and qualified pipeline for Azure, Microsoft 365, Power Platform, MS Fabric and Azure DevOps solutions, averaging ~15 qualified meetings per month.
- Led technical-consultative conversations translating business needs into Microsoft Cloud and AI solution roadmaps.

Experience — continued

Software Engineer

Innovative S.A. (Novacomp Group) | October 2022 - September 2025

- NovaMP SINPE — Electronic Payments Platform (BCCR): built configuration and financial-reporting modules and delivered hotfixes sustaining 24/7 uptime for 10+ entities certified by the Central Bank of Costa Rica.
- Reduced a high-volume reporting query from 47 s to 3.1 s through execution-plan analysis, covering indexes with INCLUDE and set-based replacements for cursor logic.
- NovaToken — Two-Factor Authentication (2FA): developed the React BackOffice and maintained OTP token logic (60-second expiry), meeting financial-regulator requirements across LATAM.
- NovaBank — Omnichannel Internet Banking: built fund-transfer modules and core-banking integrations for Coopeguanacaste and Coope San Ramón, OWASP Top 10 compliant.
- JUPEMA — Digital Case Management: delivered a new case-management and reporting system for Costa Rica's National Teachers' Pension Fund (Java · .NET · Blazor · Oracle).

Software Development Intern

Innovative S.A. | May 2022 - October 2022

- Developed an employee web and mobile app and its backoffice with Vue.js, React Native, .NET Core and SQL Server.
- Integrated OneSignal, Firebase Analytics and Crashlytics for notifications and error monitoring.
- Analyzed usage, performance and security data with Python and Power BI.

Selected Projects

AI NOVA — Multi-agent software engineering

Collaborative AI NOVA program project: product, architecture, development, security, testing and review agents in a controlled, observable workflow with guardrails. LangGraph, RAG, MCP, Langfuse, Ollama, FastAPI and React.

Docker Coding Agent

Coding-agent project using Claude/Codex in Docker Sandboxes: bounded execution, verification and reports. Public repository with documented V1 scope and limitations.

BusinessAI Analytics

Local BI with dashboards, forecasts and a bilingual chatbot. Five Spring Boot services behind Spring Cloud Gateway, React/TypeScript, FastAPI and MySQL; approximately 180 JUnit 5 tests, jqwik, pytest and Vitest, with GitHub Actions and SonarQube.

InterviewCleanAPI

.NET 10 with four-layer Clean Architecture, EF Core/MySQL, JWT and RBAC. 20 xUnit API tests using WebApplicationFactory and Selenium tests for React, Angular and Vue clients.

Education

Licentiate in Computer Engineering, Software Development emphasis (post-bachelor degree)

ULACIT | 2023–2024

Bachelor’s degree in Computer Systems Engineering

Universidad Latina de Costa Rica | 2019–2022

Certifications

Microsoft AZ-900: Azure Fundamentals · Microsoft MS-900: Microsoft 365 Fundamentals · Scrum Foundation Professional Certificate (SFPC). DP-800: SQL AI Developer Associate — in preparation, not an earned certification. Also in preparation: Anthropic, Oracle AI and Microsoft AI (Copilot) certifications.

Languages

Native Spanish · Advanced English (TOEIC 935/990)